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CARGO PACK FOR MOUNTING ON A BICYCLE

Abstract

A cargo pack for mounting on a handlebar **12** of a bicycle comprises a support in the form of a cage **20** which is connected to the handlebar **12** by arms **18.** The arms **18** are securely fixed at openings **22** to the handlebar **12** in an adjustable manner, and the cage **20** is connected to the arms **18** by respective connecting structures **38** which enable adjustment of the spacing between the arms **18** in a direction parallel to the axis of the openings **22** and which enable pivoting of the support **20** relatively to **10** the arms **18** about the axis of the openings **22,** the connecting structures **38** each having a locking mechanism for locking the support **20** with respect to the arms **18** in a selected pivoted position.

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Background/Summary

[0001] This invention relates to a cargo pack, and is particularly, although not exclusively, concerned with a cargo pack to be mounted on a bicycle, for example on the bicycle handlebars.
[0002] Bikepacking is a popular leisure pursuit which often involves multi-day touring on a bicycle, often an off-road bicycle such as a mountain bike. Everything needed for touring is carried on the bicycle or by the rider. There is therefore a need for luggage containers, often referred to as cargo packs, which can be attached to the bicycle and can be loaded with equipment such as clothing, sleeping bags and food.
[0003] “Soft” packs, such as collapsible bags, are known for this purpose, but pose particular problems. Different bicycles have widely different configurations, and in particular bicycle handlebars can vary considerably. Cargo packs intended for mounting on bicycle handlebars consequently need to be versatile in order to fit different bicycles.
[0004] When such packs are mounted on the bicycle handlebars, it is important for them to be kept away from the front wheel to avoid abrasion and the accompanying buzzing noise. The packs must also be kept away from control elements such as brake levers, gear change controls, trip computers, etc. Also, it is desirable for the packs to be centralised on the handlebars to avoid imbalance. These requirements can be difficult to achieve if the contents of the pack may vary in volume, for example if an item of clothing is taken out of, or placed in, the pack.
[0005] According to one aspect of the present invention there is provided a cargo pack carrier for mounting on a handlebar of a bicycle, the carrier comprising a support for receiving a cargo pack, and a pair of arms each attached at one end to the support and having at their other ends aligned openings for receiving the handlebar, the arms being connected to the support by respective connecting structures which enable adjustment of the spacing between the arms in a direction parallel to the axis of the openings and which enable pivoting of the support relatively to the arms about an axis parallel to the axis of the openings, the connecting structures each having a locking mechanism for locking the support with respect to the arms in a selected pivoted position.
[0006] Each connecting structure may comprise a fastener which engages a respective slot in the support. Each connecting structure may comprise a barrel bolt accommodated within a bore provided in the respective arm.
[0007] The locking mechanism of each connecting structure may be operable to fix the barrel bolt with respect to the bore. The locking mechanism may comprise the fastener of the connecting structure.
[0008] The support may have a central spine and lateral rails which are spaced from the spine. The arms may be secured to the spine.
[0009] The rails may have slots for receiving retaining straps.
[0010] The support may be provided with a support for an accessory.
[0011] In accordance with a second aspect of the present invention, there is provided a cargo pack for attachment to a bicycle handlebar, the pack comprising a cargo pack carrier as defined above and a cargo container secured to the support.
[0012] The cargo container may be secured to the support by straps. Where the support is provided with slots, the straps may be accommodated in the slots. The straps may extend around the container.
[0013] The container may be a bag.
[0014] The container may be provided with at least one hook for engagement with the support.
[0015] Another aspect of the present invention provides a cargo pack for attachment to a bicycle handlebar, the pack comprising a cargo container and a cargo pack carrier having a support to which the container is secured by at least one releasable retaining device, the container being

provided with at least one hook which engages the support to support the container on the support independently of the retaining device.
[0016] The hook may be secured to the container by welding. The hook is made from thermoplastics material and the container is made from thermoplastics sheet material.

Description

[0017] For a better understanding of the present invention, and to show more clearly how it may be carried into effect, reference will now be made, by way of example, to the accompanying drawings, in which:
[0018] FIG. **1** shows a bicycle provided with a cargo pack carrier;
[0019] FIG. **2** is a partial view corresponding to FIG. **1**, showing a cargo container fitted to the cargo pack carrier;
[0020] FIG. **3** shows the cargo pack carrier detached from the bicycle;
[0021] FIG. **4** corresponds to FIG. **3** but shows the cargo pack carrier in a different configuration;
[0022] FIG. **5** is a side view of the cargo pack carrier;
[0023] FIG. **6** corresponds to FIG. **5** but shows the cargo pack carrier in a different configuration;
[0024] FIG. **7** is a sectional view corresponding to FIG. **5**;
[0025] FIG. **8** is a sectional view corresponding to FIG. **6**
[0026] FIG. **9** shows an initial stage in the mounting of the cargo container on the cargo pack carrier;
[0027] FIG. **10** corresponds to FIG. **9** but shows a final stage in the mounting of the cargo container; and
[0028] FIG. **11** shows the cargo pack carrier fitted with retaining straps and an accessory mounting attachment.
[0029] The bicycle shown in diagrammatic form in FIGS. **1** and **2** comprises a frame **2**, front and rear wheels **4** and **6**, a chain drive **8**, a saddle **10** and a handlebar **12** controlling the front wheel **4**. The frame **2** includes a head tube **3** which supports the front wheel **4**.
[0030] The handlebar **12** comprises a tubular member having a circular cylindrical profile, although it will be appreciated that other forms of member, and other profiles, could be used. In this specification the expression “handlebar” is used to denote the entire tubular member extending from one handgrip to the other.
[0031] A cargo pack in the form of a bag **14** is supported on the handlebars **12** by a cargo pack carrier **16** shown in more detail in FIGS. **3** to **7**. The cargo pack carrier comprises a pair of arms **18** and a support in the form of a cage **20** which is supported on the handlebar **12** by the arms **18**. For this purpose, each arm **18** has an opening **22** at one end which receives the handlebar **12**. As shown in FIGS. **3** and **4**, the openings **22** of the two arms **18** are aligned with each other so as to fit on a straight handlebar **12**, although it will be appreciated that slight deviations from precise straightness may occur.
[0032] As can be appreciated from FIG. **7**, each arm **18** has a main portion **24** and a clamping portion **26** which is connected to the main portion **24** at a pivot **28**. A clamping screw **30** extends through a bore **32** in the main portion **24** into a tapped hole **34** in the clamping portion **26** to enable the arm **18** to be fitted over the handlebar **12** by swinging the clamping portion **26** outwardly and subsequently securing it by means of the clamping screw **30** to grip the handlebar **12** firmly so as to resist displacement of the arm **18** around or along the handlebar **12**. A liner **34**, for example of very stiff material such as aluminium, is provided on the internal surface of the opening **22** to enhance the grip between the arm **18** and the handlebar **12** while avoiding damage to the handlebar **12**. The liner **34** is in two parts, corresponding to the main portion **24** and the clamping portion **26**.
[0033] At the end of each arm **18** away from the opening **22**, the arm is provided with a connecting

structure **36** by which the cage **20** is adjustably fitted to the arm **18**. The connecting structure **36** comprises a barrel bolt **38** fitted within a plain bore **40** formed in the main portion **24** of the arm **18**. The barrel bolt **38** has a tapped cross-bore **42** which receives a fastener **43**. The arm **18** has a circumferential slot **44** which extends part of the way around the bore **40** receiving the fastener **43** when engaged in the tapped cross-bore **42**.

[0034] The cage **20** is made from any suitable substantially rigid material, such as a metallic alloy or a rigid plastics material. It comprises a central spine **46** and a pair of lateral rails **48**, **50**. As can be appreciated from FIG. **10**, the upper rail **48** is straight, whereas the lower rail **50** is kinked, so as to provide an upward displaced region **52**. The central spine **46** is connected to the upper and lower rails **48**, **50** by relatively thin connecting sections **54**. The cage **20** is thus of a relatively light-weight structure.

[0035] As can be appreciated from FIGS. **5** and **6**, the cage **20** has a convex shape as viewed looking towards the arm **18** so as to serve as a cradle extending partially around the bag **14**. The central spine **46** is provided with a pair of counterbored slots **56** positioned to receive the fasteners **43** engaged with the cross-bores **42** in the barrel bolts **38**. It will be appreciated that the slots **56** extend parallel to the common axis of the openings **22** and thus allow for the arms **18** to be fixed to the cage **20** at a variety of positions along the central spine **46**, so enabling the spacing between the arms **18** to be selected so as to fit in desired positions on the handlebar **12**.

[0036] The lateral rails **48**, **50** are provided with further slots **58** which, as shown in FIG. **10**, can receive retaining elements such as Voile Straps® **60** for securing the bag **14** to the cage **20**.

[0037] Additional counterbored slots **62** are provided in the upper rail **48** for the attachment of accessories such as GPS navigation aids, lights, cameras or mobile phones, for example by way of mountings **64** which may be secured to the cage **20** by suitable fasteners **65** extending through the slots **62**.

[0038] FIGS. **8** and **9** show the cargo pack carrier **16** fitted with a cargo pack **14** in the form of a container such as a bag. Although represented diagrammatically as a cylinder, the bag **14** may be made from a flexible material, such as a waterproof textile or sheet material, for example a 420D nylon or other plastics material, which is sufficiently stiff to maintain its shape when unstressed, but which will flex if, for example, baggage items are pressed forcefully into the bag, or if straps or the like are tensioned around the periphery of the bag. The bag **14** has an opening (not shown) for loading or unloading the bag and is also provided with a reinforcing layer **66** of an abrasion-resistant material such as thermoplastic polyurethane (TPU). The layer **66** may, for example, be bonded to the material of the bag **14** by welding.

[0039] The bag **14** is also provided with a pair of spaced-apart hooks **68** which may also be made from TPU and secured to the bag **14** by welding. In the embodiment shown in FIGS. **8** and **9**, the hooks **68** are shown as secured to the reinforcing layer **66**, they may be independently fixed to the material of the bag **14**, with the reinforcing layer being disposed at a different position on the back **14**, for example in the region of the bag **14** most likely to come into accidental contact with the front wheel **4**.

[0040] As shown in FIGS. **8** and **9**, the hooks **68** are disposed on the bag **14** so that they can be placed over the upper rail **48**. Thus, in use, the bag **14** can initially be suspended from the cage **20** by means of the hooks **68**. This enables the user to fasten the straps **60**, which may require the use of both hands, without at the same time leading to support the bag **14**.

[0041] For use, the arms **18** are fitted to the cage **20** by the fasteners **43** passed through the slots **56** into the cross-bores **42** in the barrel bolts **38**. Initially, the fasteners **43** are not fully tightened, so that the arms **18** can be displaced laterally along the slots **56**, and also pivoted with respect to the cage **20** by allowing the barrel bolt **38** to rotate within the bore **40**, as permitted by the slot **44**.

[0042] The arms **18** are then fitted over the handlebar **12** by unscrewing the clamping screws **30** of each arm **18** so that the clamping portion **26** can be swung about the pivot **28** to allow the arm **18** to be placed over the handlebar **12**. The clamping portion **26** is then closed and the arm **18** is fastened

in a desired position along the handlebar **12** by tightening the clamping screw **30**. When the cage **20** as reached it's a desired position, both about the axis of the handlebar **12**, and about the axis of the barrel bolts **38**, the clamping screws **30** can be fully tightened, as can the fasteners **43** in the cross-bores **42**. This fixes the position of the cage **20** with respect to the handlebar **12**. By way of example, FIGS. **5** and **6** illustrate two alternative positions for the cage **20**. In FIG. **5**, the cage **20** holds the bag **14** at a forward facing position with respect to the handlebar **12** whereas FIG. **6** shows the cage **20** in a downwardly facing position. It will be appreciated that the cage **20** can be fixed in any orientation between the two positions shown in FIGS. **5** and **6**.
[0043] In order to mount the bag **14**, it is first fitted onto the cage **20** by placing the hooks **66** over the upper rail **48** of the cage **20**, leaving the user's hands free to connect up and tighten the straps **60**. The straps **60** are then stretched around the bag **14** and secured by means of buckles.
[0044] The structure of the cargo pack carrier **16** thus enables the positioning of the bag **14** suitably on the bicycle with a wide degree of versatility to select the bag position, taking account of its size and contents so as to avoid contact with the front wheel four, and to minimise intrusion into the movement envelope of the rider. Also, the structure of the cage **20**, and in particular the slots **62**, enable additional accessories to be mounted.
[0045] The features of the present invention provide a versatile cargo pack carrier while minimising the component count and the operations required to assemble and fit the cargo pack carrier. Also, the fitting of a bag to the cargo pack carrier is simplified by enabling the bag to be supported on the cage **20** by the hooks **66** while final securing by means of the straps **60** takes place.

Claims

1. A cargo pack carrier for mounting on a handlebar of a bicycle, the carrier comprising a support for receiving a cargo pack, and a pair of arms each attached at one end to the support and having at their other ends aligned openings for receiving the handlebar, the arms being connected to the support by respective connecting structures which enable adjustment of the spacing between the arms in a direction parallel to the axis of the openings and which enable pivoting of the support relatively to the arms about an axis parallel to the axis of the openings, the connecting structures each having a locking mechanism for locking the support with respect to the arms in a selected pivoted position.
2. A cargo pack carrier as claimed in claim 1, in which each connecting structure comprises a fastener which engages a respective slot in the support.
3. A cargo pack carrier as claimed in claim 1, in which each connecting structure comprises a barrel bolt accommodated within a bore provided in the respective arm.
4. A cargo pack carrier as claimed in claim 3, in which the locking mechanism of each connecting structure is operable to fix the barrel bolt with respect to the bore.
5. A cargo pack carrier as claimed in claim 2, in which the locking mechanism comprises the fastener of the connecting structure.
6. A cargo pack carrier as claimed in claim 1, in which the support has a central spine and lateral rails which are spaced from the spine.
7. A cargo pack carrier as claimed in claim 6, in which the arms are secured to the spine.
8. A cargo pack carrier as claimed in claim 6, in which the rails have slots for receiving retaining straps.
9. A cargo pack carrier as claimed in claim 1, in which the support is provided with a support for an accessory.
10. A cargo pack for attachment to a bicycle handlebar, the pack comprising: a cargo pack carrier comprising cargo pack carrier for mounting on a handlebar of a bicycle, the carrier comprising a support for receiving a cargo pack, and a pair of arms each attached at one end to the support and

having at their other ends aligned openings for receiving the handlebar, the arms being connected to the support by respective connecting structures which enable adjustment of the spacing between the arms in a direction parallel to the axis of the openings and which enable pivoting of the support relatively to the arms about an axis parallel to the axis of the openings, the connecting structures each having a locking mechanism for locking the support with respect to the arms in a selected pivoted position; and a cargo container secured to the support.

11. A cargo pack as claimed in claim 10, in which the cargo container is secured to the support by straps.

12. A cargo pack as claimed in claim 11, in which the support of the cargo pack carrier has a central spine and lateral rails which are spaced from the spine, and in which the rails of the cargo pack carrier have slots for receiving retaining straps, and in which the straps are accommodated in the slots.

13. A cargo pack as claimed in claim 12, in which the straps extend around the container.

14. A cargo pack as claimed in claim 10, in which the container is a bag.

15. A cargo pack as claimed in claim 10, in which the container is provided with at least one hook for engagement with the support.

16. A cargo pack as claimed in claim 15, in which the hook is secured to the container by welding.

17. A cargo pack as claimed in claim 16, in which the hook is made from thermoplastics material and the container is made from thermoplastics sheet material.

18. A cargo pack for attachment to a bicycle handlebar, the pack comprising a cargo container and a cargo pack carrier having a support to which the container is secured by at least one releasable retaining device, the container being provided with at least one hook which engages the support to support the container on the support independently of the retaining device.

19. A cargo pack as claimed in claim 18, in which the hook is secured to the container by welding.

20. A cargo pack as claimed in claim 19, in which the hook is made from thermoplastics material and the container is made from thermoplastics sheet material.